PRACTICAL COURSE WORKBOOK

Deep Learning & Neural Networks

Turn untidy data into a defensible finding

LEVEL	Advanced
GUIDED TIME	5 hours across 12 lessons
PROJECT	a cleaned dataset, analysis and decision-ready chart
SKILLS	PyTorch, TensorFlow, CNN, Transformers, Critical evaluation, Documentation
EDITORIAL STATUS	Structured draft v0.9

WHAT YOU WILL BE ABLE TO DO

- Use PyTorch appropriately in a realistic, bounded task.
- Use TensorFlow appropriately in a realistic, bounded task.
- Use CNN appropriately in a realistic, bounded task.
- Use Transformers appropriately in a realistic, bounded task.

WHO THIS IS FOR

Learners using data and code to answer practical questions.

BEFORE YOU BEGIN

- Basic numeracy and file-management skills

COURSE MAP

From foundation to finished work

Deep Learning & Neural Networks is a structured, practical course covering PyTorch, TensorFlow, CNN, Transformers. It emphasizes explainable methods, checked work and a final outcome that can be reviewed.

MODULE	LESSONS AND FOCUS
01 PyTorch: Data foundations	1. Problem framing with PyTorch 2. Data types with TensorFlow 3. Tool setup with CNN
02 TensorFlow: Analysis	1. Cleaning with Transformers 2. Exploration with PyTorch 3. Modelling with TensorFlow
03 CNN: Evaluation	1. Validation with CNN 2. Uncertainty with Transformers 3. Communication with PyTorch
04 Transformers: Applied project	1. Workflow with TensorFlow 2. Review with CNN 3. Reproducibility with Transformers

HOW TO STUDY EACH LESSON

1. Read the objective and identify the decision it supports.
2. Reproduce the worked example with the stated input.
3. Test one normal case and one boundary or failure case.
4. Complete the knowledge check without guessing.
5. Save a short evidence note: result, limitation and next action.

CAPSTONE PROJECT

Produce a reviewable Deep Learning & Neural Networks project using PyTorch, TensorFlow, CNN.

DELIVERABLE	A working Deep Learning & Neural Networks artefact plus an evidence-based self-review.
TOOLS	A suitable PyTorch environment, A plain-text decision log, Test data or realistic sample material

Production plan

1. Define the intended user, outcome and constraints.
2. Create the smallest complete result using PyTorch.
3. Test one normal case, one boundary case and one failure response.
4. Revise the work from the evidence and preserve before-and-after results.
5. Prepare a concise handover containing method, limitations and next step.

Definition of done

- The output matches the stated outcome.
- Inputs and decisions are reproducible.
- Boundary and failure evidence is included.
- Limitations and responsibility considerations are explicit.

Evidence log

INPUT	Original brief, data, requirements or test material.
DECISIONS	Chosen method and one reasonable alternative.
TESTS	Normal, boundary and failure results.
LIMITATION	What the result does not establish.
REVISION	One evidence-led improvement.

REVIEW AND SOURCES

Make the work credible and repeatable

Final self-review

- Can another learner repeat the method?
- Did I test a difficult case?
- Did I avoid unsupported claims?
- Is the next action proportionate to the remaining risk?

Authoritative references

The Python Tutorial

Python Software Foundation - reviewed 2026-08-21
https://docs.python.org/3/tutorial/

User Guide

scikit-learn - reviewed 2026-08-21
https://scikit-learn.org/stable/user_guide.html

SOURCE NOTE

Product versions, laws and professional standards can change. Recheck the linked primary source before applying version-sensitive information.

NEXT IMPROVEMENT

Choose one weakness found during review and improve it before extending the PyTorch scope.